

Lima v2.2

Now supports
Mac & Win
guests

PromCon

North America 2021

Linux virtual machines

for running containers, sandboxing AI agents, etc.

- Automatic host filesystem sharing
- Automatic port forwarding
- Built-in integration for containerd

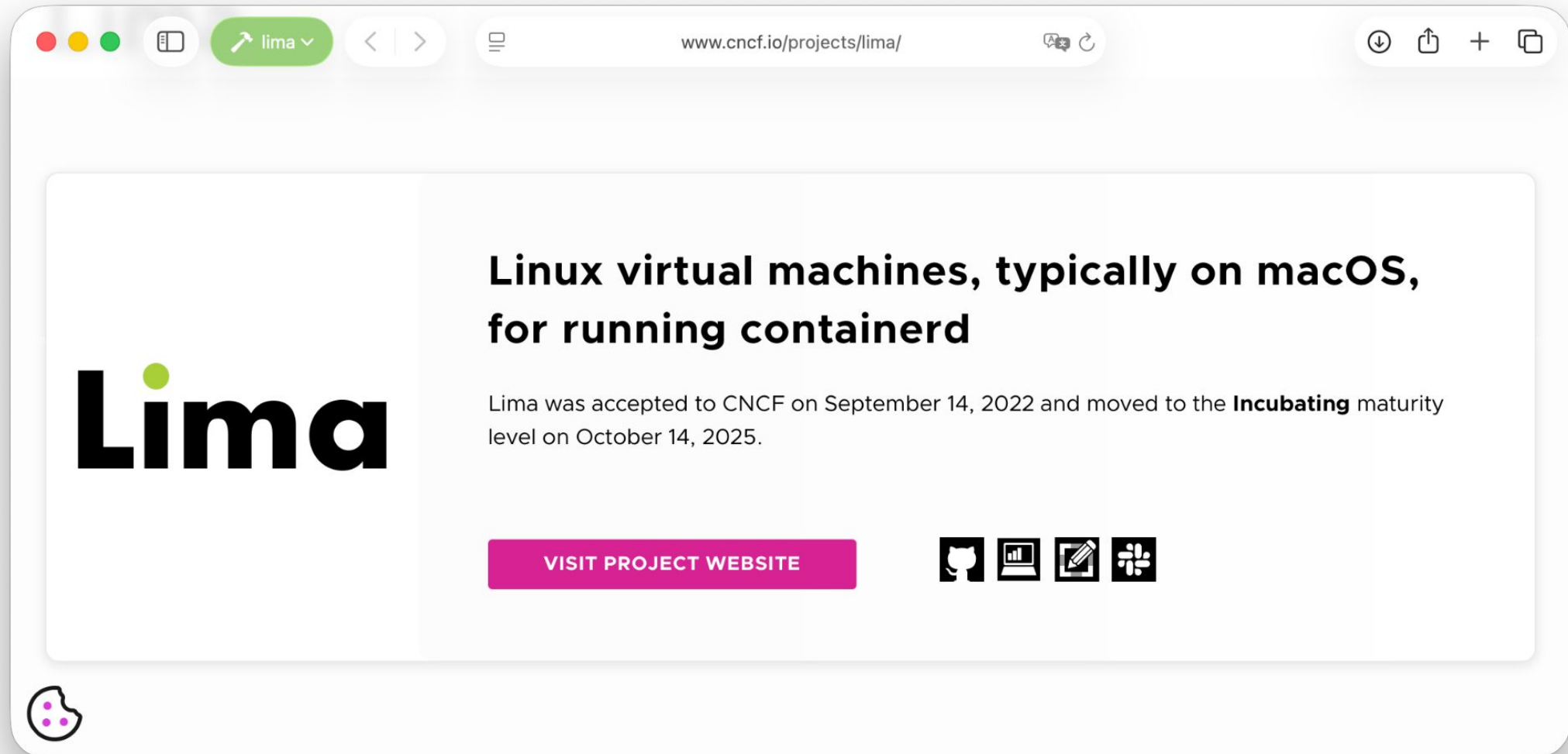
(and Docker, Podman, Kubernetes, Apptainer, ...)

```
$ brew install lima  
$ limactl start  
$ lima nerdctl run -p 80:80 nginx
```

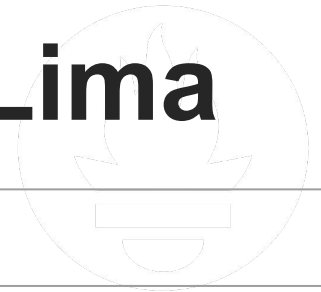


<https://lima-vm.io/>

Promoted to CNCF Incubating Project



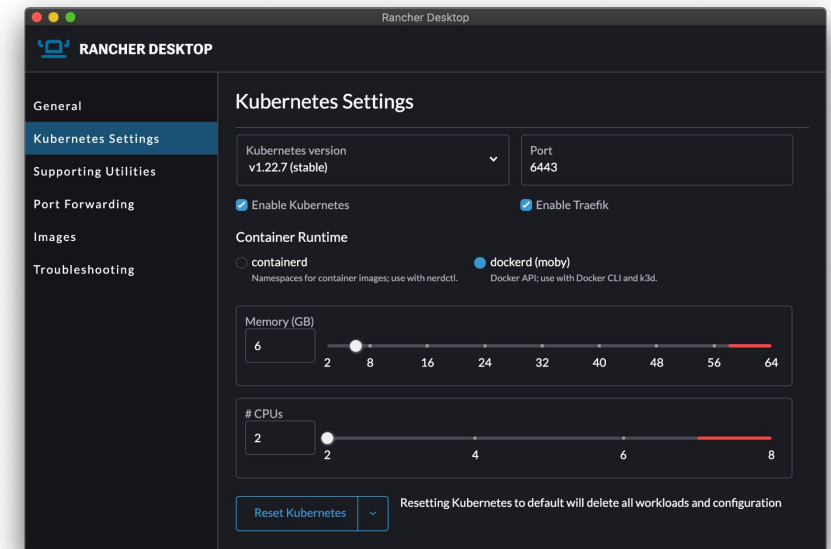
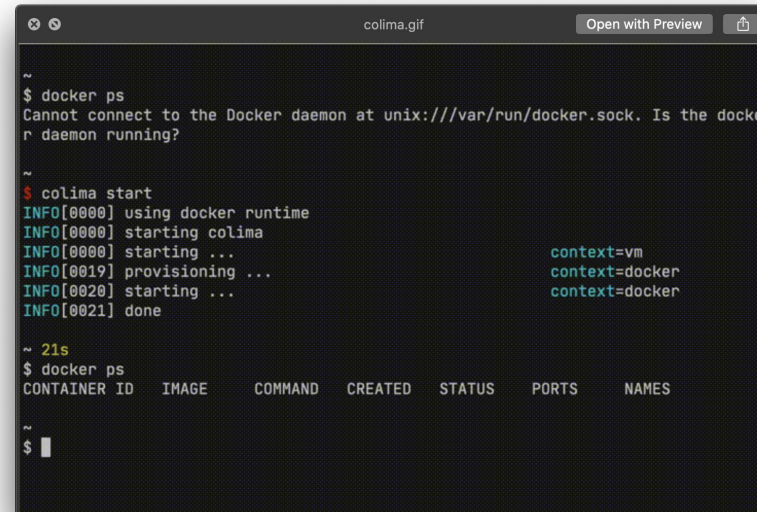
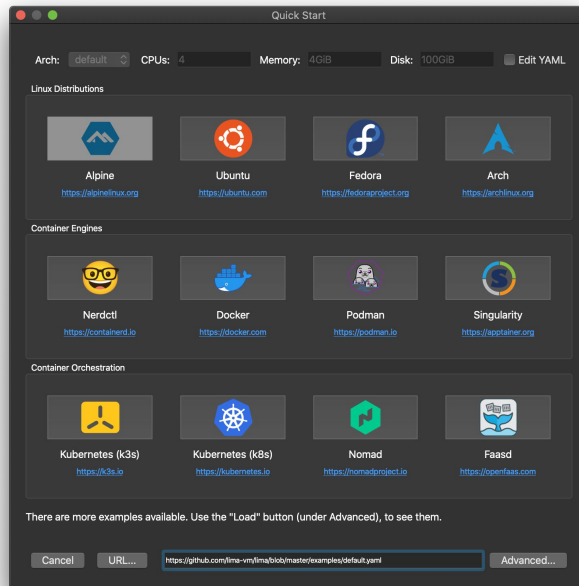
Third party FLOSS projects based on Lima



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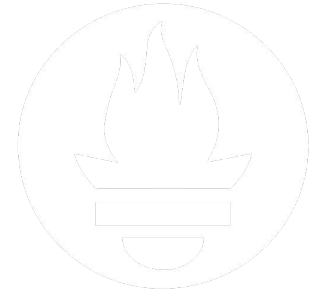
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Lima-GUI	https://github.com/afbjorklund/lima-gui
Colima	https://github.com/abiosoft/colima
Rancher Desktop	https://rancherdesktop.io/
Finch	https://github.com/runfinch/finch
Podman Desktop	https://podman-desktop.io/



v2.2 (July 2026)

- Windows Server 2025 guests
- Windows 11 guests

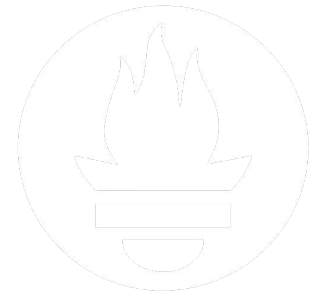


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v2.1 (March 2026)

- macOS guests
- FreeBSD guests
- `--sync` to prevent AI agents from breaking the host files

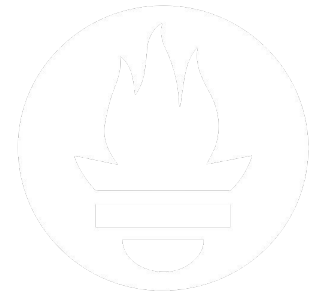
```
$ limactl start --mount-none
$ lima --sync . claude "Implement something"
[...]
⚠ Accept the changes? (added: 2, deleted: 1, modified: 3, ...)
→ Yes
  No
  View the changed contents
```



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v2.0 (November 2025)

- Plugin infrastructure to allow implementing new features without modifying Lima
 - VM driver plugins
 - CLI plugins
 - URL schema plugins
- GPU acceleration with krunkit VM driver
- MCP server for protecting AI agents such as Gemini
- Lots of CLI improvements



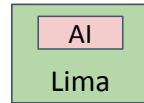
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Extending the focus to AI

- Original goal in 2021 was to facilitate running containers on macOS

- Turned out to be highly useful for securing AI agents too

- AI inside Lima



- Just run Codex, Copilot, Claude, Gemini, etc. inside Lima
- LLM inference can be done inside Lima, using GPU acceleration

- AI outside Lima

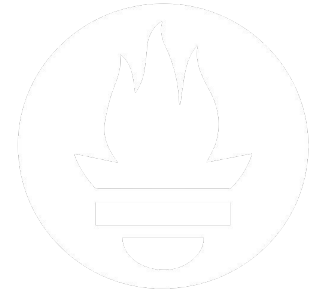


- Lima's MCP server can be connected from AI agents running on the host
- VScode + Remote SSH + Copilot works well too with Lima



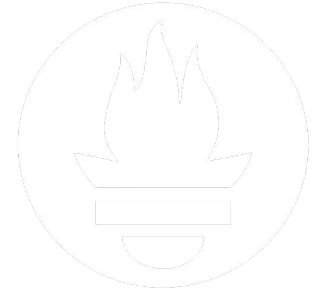
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GPU acceleration in action



```
lima
$ lima --version
limactl version 2.0.0
$ lima llama-cli --version
ggml_vulkan: Found 1 Vulkan devices:
ggml_vulkan: 0 = Virtio-GPU Venus (Apple M4 Max) (venus) | uma: 1 | fp16: 1 | bf
16: 0 | warp size: 32 | shared memory: 32768 | int dot: 0 | matrix cores: none
version: 6962 (230d1169e)
built with cc (GCC) 15.2.1 20251022 (Red Hat 15.2.1-3) for aarch64-redhat-linux
$
```


How it works



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- **Hypervisor**

Virtualization.framework aka vz, QEMU, or krunkit

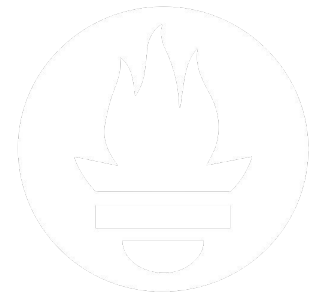
- **Intel-on-ARM binary executor**

Rosetta 2 (vz) or QEMU-user

- **Filesystem sharing**

virtiofs (vz, krunkit), virtio-9p (QEMU), or reverse-sshfs

How it works



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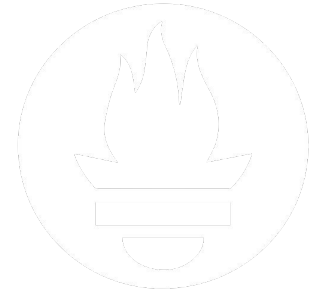
- **Network**

- user mode networking (default)
- `socket_vmnet` (for direct access to the IP, etc. with sudo)
- `vzNAT` (for faster direct access to the IP, with vz)

- **Port forwarding**

- `NETLINK_SOCK_DIAG` watcher (for most services)
- Kubernetes service watcher (for Kubernetes services)

Built-in templates



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- **Distros**

almalinux, alpine, archlinux, centos-stream, debian, opensuse, oraclelinux, rocky, ubuntu, ...

- **Container engines**

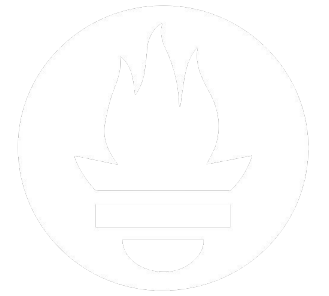
apptainer, docker, docker-rootful, podman, podman-rootful, ...

- **Container orchestration**

faasd, k3s, k8s

```
$ limctl start --name=default template://docker
```

Future ideas



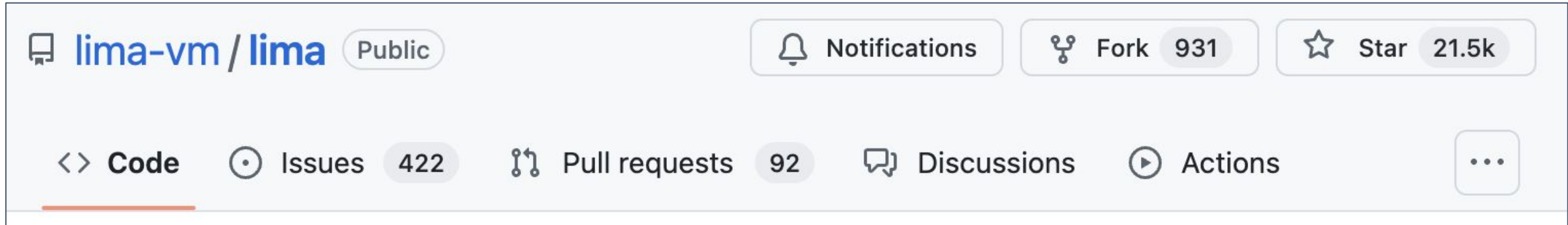
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- More VM drivers
 - Based on HCS (Host Compute System) for Windows hosts
 - e.g., for managing IaaS instances
- Menu-based text user interface
- UX improvement for composing multiple VMs
- Support for AI workloads inside VM on Linux and Windows host
- Packet filtering for AI agents

Join the community!

- **Web site:** <https://lima-vm.io/>

- **GitHub:** <https://github.com/lima-vm/lima>



- **Slack:** <https://slack.cncf.io/> (Channel: #lima)